EMILLY RAIANE RODRIGUES

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PROFILE SUMMARY

Dual-degree student in Aerospace Engineering and Embedded Systems, working on flight software for satellite systems across research, space agencies, and Brazilian industry. Focused on building reliable real-time embedded systems under strict constraints. Interested in turning complex space technologies into scalable and accessible solutions, with experience leading initiatives that expand access to aerospace.

PROFESSIONAL EXPERIENCE

Machine Learning & FPGA Research Intern **July 2026-Present**
European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

- Selected as one of 47 students from more than 7,000 candidates for the CERN openlab Summer Student Programme.
- Research and development of an FPGA-targeted autoencoder for processing ATLAS calorimeter data and reconstructing jets in real time.

Embedded Flight Software Intern **October 2025 - June 2026**
Visiona Tecnologia Espacial S.A. *São José dos Campos, Brazil*

- Nanosatellite flight software development, focusing on on-board data handling (OBDH).
- Validation of telecommands and telemetry from the ground to ensure correctness, reliability, and proper integration with the satellite systems.
- Development of the camera module, defining operating modes and integrating the firmware in collaboration with the optical team.

Embedded Flight Software Intern **February 2025 - August 2025**
Centre national d'études spatiales (CNES) *Toulouse, France*

- Developed flight software for a satellite system in a real-time, resource-constrained environment requiring high reliability and fault tolerance.
- Designed a recovery mechanism restoring system state from FRAM, reducing downtime from 7 hours to minutes and improving mission readiness.
- Redesigned task scheduling to prioritize critical memory, reducing startup time from 40+ seconds to under 5 seconds
- Implementation of satellite operating modes, automatically setting equipment to power on and off according to each mode, ensuring energy savings.

Embedded Flight Software Intern **May 2024 - August 2024**
TIMA Laboratory *Grenoble, France*

- Developed communication between an STM32 and a Raspberry Pi, improving test tracking to make it faster and more reliable.
- Reduced total execution time from 8.6 seconds to 0.014 seconds by eliminating manual synchronization and enabling state recovery.
- Developed automated Bash tools to control tests, handle errors, and collect results during radiation experiments on a particle accelerator.
- Designed and validated error detection mechanisms for communication faults induced by radiation and electromagnetic interference.

Satellite Control Researcher **September 2022 - September 2023**
Instituto Tecnológico de Aeronáutica (ITA) *São José dos Campos, Brazil (Remote)*

- Contributed to the ITASAT-2 mission, which focuses on space weather research and the development of advanced satellite technologies.
- Developed algorithms in MATLAB to estimate satellite position from noisy sensor data, improving accuracy by 82%.
- Designed control systems to stabilize the satellite and ensure reliable operation.

Machine Learning Intern **February 2022 - September 2022**
Dafiti Group *São Paulo, Brazil (Remote)*

- Implemented computer vision algorithms in Python for 2D image adjustment and image segmentation in a virtual clothing try-on application.
- Segmented human figures and clothing items to place clothes correctly on images.
- Detected that the training dataset was not representative of Brazilian customers and started retraining the U-Net segmentation network.
- Started moving the model from PyTorch to TensorFlow to use TPU for faster training.

EDUCATION

Master M2 in Embedded Systems (Dual Degree Program) **September 2023 - August 2025**
Institut Polytechnique de Grenoble (Grenoble INP - Phelma) *Grenoble, France*

- Recipient of a Brazilian government scholarship for a dual degree program in France due to academic excellence.

Bachelor in Aerospace Engineering **March 2020 - December 2026** ***(Expected Graduation)***
Universidade Federal de Santa Maria (UFSM) *Santa Maria, Brazil*

- One of 43 students selected from more than 11,000 candidates for the Líderes Estudar scholarship, which recognizing Brazil's top university students.
- Awarded a national prize recognizing the top university students in Brazil, selected as one of 6 recipients from more than 8,000 candidates.
- Selected as one of 10 top Brazilian students for the fully funded conference in Sunnyvale, held at Google focused on innovation and entrepreneurship.
- Speaker on two panels on the new space economy (200+ attendees each) at South Summit, a global event with 23,000+ participants and 900+ investors.

PROJECTS

Validation of 128-bit CVA6 Processor **September 2024 - January 2025**

- Validated the 128-bit extension of the CVA6 RISC-V processor, ensuring correctness across 32-bit, 64-bit, and newly introduced 128-bit instructions
- Developed and automated instruction-level test suites in Assembly and Python to verify arithmetic, load/store, and sign-extension behaviors.
- Identified and debugged a cache-related memory access issue caused by incorrect load/store offset handling after extending the data width to 128 bits, leading to misaligned address calculations.
- Analyzed simulation logs and waveforms to trace register-level behavior and validate bit-level transitions over time using Verilog and SystemVerilog
- Debugged and validated memory access logic by correlating ABI register usage with internal register files and waveform representations.

Fault Injection in Memory **September 2024 - November 2024**

- Injected Single Event Upset (SEU) faults to evaluate and enhance fault tolerance in systems, ensuring reliability in high-radiation environments.
- Analysis of critical points (FSM, registers, and arithmetic units) and memory mapping for fault injection.
- Development of VHDL code, set up a Design Under Test (DUT) and testbench for to simulate fault injection.
- Automation of simulations with TCL scripts, covering all fault points and evaluating performance impact.
- Comparison of execution times with MTBF (Mean Time Between Failures) and analysis of the impact of faults on the system.

Design of a Digital FIR (Finite Impulse Response) Filter **January 2024 - May 2024**

- Specification, implementation, and validation of a digital FIR filter in VHDL for signal processing.
- Implemented abstract ADC and DAC interfaces to manage AD7822 (ADC) and AD7302 (DAC) converters.
- Used ModelSim and TCL scripts to automate the simulation process.
- Prototyping on FPGA using the Spartan 3 kit, with results analysis through an oscilloscope.

Design of Digital Components **September 2023 - December 2024**

- Developed a VHDL memory bank from a register, integrating typical memory access protocol.
- Created a testbench to generate test stimuli and analyze waveforms before and after synthesis.
- Used ModelSim and TCL scripts to automate the simulation process.
- Specified and implemented a finite state machine (FSM) to detect the start of an Ethernet packet.

Development of an Image Filter on System on Chip **September 2024 - December 2024**

- Designed an image filter using C++ and SystemC, leveraging High-Level Synthesis (HLS) to generate FPGA-compatible RTL.
- Programmed a finite state machine (FSM) in Verilog to manage filter application to image pixels.
- Developed a testbench in C++ to generate test images and analyze waveforms.
- Simulated performance and verified functionality using transaction-level modeling (TLM).

Real-Time Embedded Software for Autonomous Vehicle **September 2024 - January 2025**

- Developed bare-metal real-time software for an autonomous vehicle controlling DC motors, steering servo, and linear camera (video demonstration)
- Designed and tuned PI controllers for each wheel to regulate speed via PWM.
- Implemented position and trajectory control algorithms to keep the vehicle centered on lines, including edge detection and corrective actuation.

Embedded Software for a CubeSat **January 2024 - May 2024**

- Support the development of a CubeSat prototype that orients solar panels based on light source detection (video demonstration).
- Developed an embedded software on ESP32 with FreeRTOS and state machine for operational mode management.
- Implemented UDP communication protocols between the satellite microcontroller and the ground station.
- Developed autonomous embedded software to manage tasks by integrating attitude control algorithms.
- Implemented an energy management system to optimize the satellite's power consumption.

Spell Checker **October 2023 - January 2024**

- Implemented advanced data structures, including tries, hash tables, and balanced binary trees.
- Organized project structure using Makefiles for compilation and execution.
- Analyzed performance and complexity to optimize processing speed and memory usage.

Development of a Low-Level OS Kernel **September 2023 - December 2023**

- Designed and implemented a low-level operating system kernel using the QEMU processor emulator.
- Implemented direct interaction with hardware registers in C using bit masking and memory-mapped IO to configure peripherals and system behavior
- Implemented UART communication in C for data exchange between the display system and the user interface.
- Developed interrupt handling and context switching in C and Assembly, configuring interrupt controller registers and managing execution state.

SKILLS

Programming: C, Python, Bash, VHDL, Assembly, MATLAB; familiar with Verilog, SystemC, C++, Ruby
Tools: OpenOCD, Xilinx SDK, Simulink, LaTeX, Git, SVN, Linux, Makefile, CMake, ModelSim, Verilator
Embedded Electronics: Microcontrollers (STM32, ESP32), SoCs (ARM Cortex-M/R, Xilinx Zynq), FPGAs (Spartan-3, Zynq-7000)
Languages: French (Advanced - C1), English (Upper-Intermediate - B2), Portuguese (Native), Spanish (Intermediate - B1)